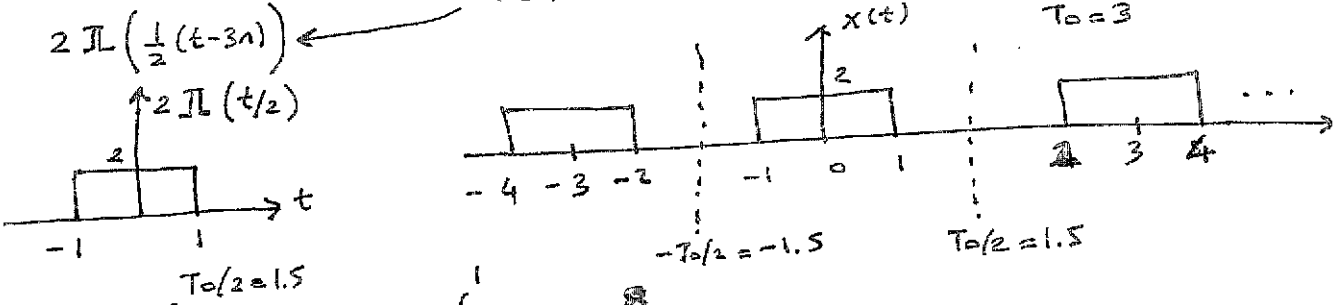
KEY

ELEC361 Quiz, 22.10.2019 Name: Number:

Question	1	2	3	4	5	6	Total
Point							

Closed books and notes. You get full or zero credit from each question. Provide the solutions and write the final answers into the boxes. Time: 40 minutes.

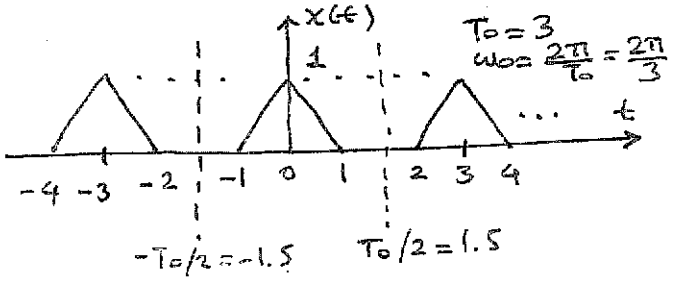
1) (10 p) If $x(t) = \sum_{n=-\infty}^{\infty} 2\prod\left(\frac{t-3n}{2}\right)$, find the power / energy of the signal $x(t)$.



$$P_x = \frac{1}{T_o}\int_{-T_o/2=-1.5}^{1} x^2(t)\,dt = \frac{1}{3}\int_{-1}^{1} 4\,dt = \frac{8}{3}$$

$$\boxed{P_x = \frac{8}{3} \text{ watt}}$$

2) (15 p.) If $x(t) = \sum_{n=-\infty}^{\infty} \Lambda(t-3n)$, find Fourier series coeffients a_n.



$$a_n = \frac{2}{T_o}\int_{-T_o/2=-1.5}^{T_o/2=1.5} x(t)\cdot\cos(n\omega_o t)\,dt$$

$$= \frac{2}{3}\cdot 2\int_0^1 (1-t)\cos\left(n\underbrace{\tfrac{2\pi}{3}}_{\omega_o} t\right)dt$$

$$= \frac{4}{3}\left[\underbrace{\frac{\sin(n\omega_o t)}{n\omega_o}\Big|_0^1}_{\sin(n\omega_o)/n\omega_o} - \int_0^1 t\cdot\cos(n\omega_o t)\,dt\right] = \frac{4}{3}\left[+\underbrace{\int_0^1 \frac{\sin(n\omega_o t)}{n\omega_o}\,dt}_{-\frac{\cos(n\omega_o t)}{(n\omega_o)^2}\Big|_0^1}\right]$$

$$= \frac{3}{(\pi n)^2}\left[1-\cos\left(n\cdot\frac{2\pi}{3}\right)\right]$$

$u = t \quad du = dt$

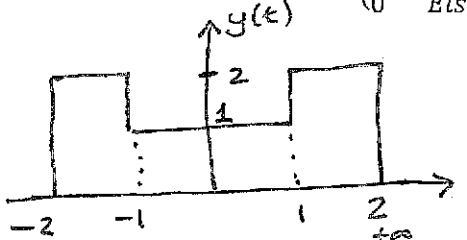
$\cos(n\omega_o t)\,dt = dv$

$v = \frac{\sin(n\omega_o t)}{n\omega_o}$

$$= \frac{4}{3}\left[\frac{\sin(n\omega_o)}{n\omega_o} - \left\{t\cdot\frac{\sin(n\omega_o t)}{n\omega_o}\Big|_0^1 - \int_0^1 \frac{\sin(n\omega_o t)}{n\omega_o}\,dt\right\}\right]$$

$$\boxed{a_n = \frac{3}{(\pi n)^2}\left[1-\cos\left(n\cdot\frac{2\pi}{3}\right)\right]}$$

3) (5 p.) $y(t) = \begin{cases} 1 & |t| \le 1 \\ 2 & 1 \le |t| \le 2 \\ 0 & \text{Else} \end{cases}$ If $Y(f) = \mathcal{F}\{y(t)\}$, find $Y(0)$.

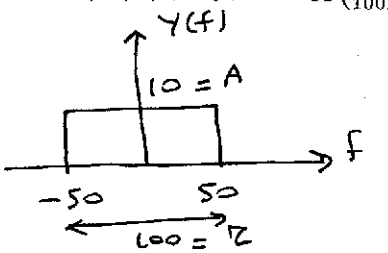


$$Y(f) = \mathcal{F}\{y(t)\} = \int_{-\infty}^{+\infty} y(t)\,e^{-j2\pi ft}\,dt \Rightarrow Y(0) = \int_{-\infty}^{+\infty} y(t)\,dt$$

$$Y(0) = \int_{-\infty}^{\infty} y(t)\,dt = 2\int_0^2 y(t)\,dt = 2\left[\int_0^1 1\,dt + \int_1^2 2\,dt\right] = 6$$

$$\boxed{Y(0) = 6}$$

4) (5 p.) $Y(f) = 10\prod\left(\frac{f}{100}\right)$. If $Y(f) = \mathcal{F}\{y(t)\}$, find $y(0)$.



$$y(t) = \mathcal{F}^{-1}\{Y(f)\} = \int_{-\infty}^{+\infty} Y(f)\cdot e^{+j2\pi ft}\,df$$

$$y(0) = \int_{-\infty}^{+\infty} Y(f)\,df = \int_{-50}^{50} 10\,df = 1000$$

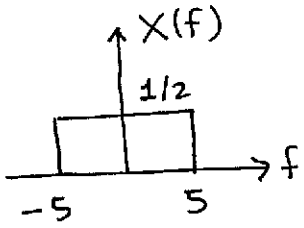
or $y(t) = A\tau\,\mathrm{Sinc}(t\tau) = 1000\,\mathrm{Sinc}(100t)$

$y(0) = 1000\,\mathrm{sinc}(0) = 1000$

$$\boxed{y(0) = 1000}$$

5) (5 p.) If $x(t) = 5Sinc(10t)$, find the energy of $x(t)$, E_x.

$$x(t) = \underset{A\tau}{5}\,\mathrm{Sinc}\,(\underset{\tau}{10}t) \longleftrightarrow X(f) = A\Pi\left(\frac{f}{\tau}\right) = \frac{1}{2}\Pi\left(\frac{f}{10}\right)$$



$$E_x = \int_{-\infty}^{+\infty} |x(t)|^2 dt = \int_{-\infty}^{+\infty} |X(f)|^2 df = \int_{-5}^{5} \left(\frac{1}{2}\right)^2 df = \frac{1}{4}\cdot 10 = 2.5\ \mathrm{J}$$

$$\boxed{E_x = 2.5\ \mathrm{J}}$$

6) (10 p.) If $x(t) = \frac{2}{1+j2\pi t}$, find the Fourier transform $X(f)$.

$e^{-at}u(t)$, $a>0$ $\overset{\mathcal{F}}{\longleftrightarrow}$ $\frac{1}{a+j2\pi f}$ Found in class.

By duality principle, $2\cdot\frac{1}{1+j2\pi t} \xrightarrow{\mathcal{F}} 2\cdot e^{+f}u(-f) = 2e^{f}$, $f\le 0$

$$\boxed{X(f) = 2e^{f}u(-f)}$$